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A Dual-Stream Convolution-GRU-Attention Network for Automatic Modulation Classification

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***Abstract*—Automatic Modulation Classification (AMC) represents a technique utilised to discern the modulation scheme employed in radio signals at the receiver's end. This holds substantial importance within intelligent receivers, which serve as pivotal components for upcoming wireless communication technologies like cognitive radio and adaptive modulation. Deep learning (DL) has gained widespread use in automatic modulation classification due to its exceptional learning capabilities and enhanced performance. In this paper, we introduce a novel hybrid dual-stream structure that combines a Convolutional Neural Network (CNN) and Gated Recurrent Unit (GRU) architecture for the classification of seven distinct digital modulation types (QAM16, CPFSK, 8PSK, QAM64, GFSK, BPSK, QPSK). Furthermore, we incorporate an attention mechanism to augment the performance of model. Notably, the utilization of two streams yields more effective outcomes compared to employing single-stream configurations. When compared with state-of-the-art approaches, our proposed model exhibits superior classification accuracy while concurrently reducing complexity. This streamlined architecture significantly enhances both training and classification efficiency.**

***Index Terms*—GRU, CNN, Modulation Classification, Attention Mechanism.**

I. Introduction

Artificial intelligence (AI) is gaining widespread popularity in the field of wireless communication, primarily driven by advanced technologies like cognitive radio and adaptive modulation. These technologies' efficiency hinges on acquiring precise information about the radio spectrum, including the utilization of frequency bands, the modulation scheme employed in radio signals, and more. Automatically determining a radio signal's modulation type at the receiver's end is essential for both military and commercial applications. Nonetheless, this task becomes significantly challenging when no prior knowledge about the transmitted data and various unknown parameters at the receiver, such as phase offsets, signal power, timing information, and carrier frequency, is available. This complexity is further compounded in real-world scenarios due to constantly changing channels and the presence of multipath fading. In commercial systems such as software-defined radios (SDR), recognizing the modulation format blindly poses a major challenge, often necessitating additional information to reconfigure the SDR system. However, the integration of intelligent receivers offers a solution by enabling the use of blind techniques, thereby enhancing transmission efficiency and reducing overhead. Over the years, several methods have been tried for AMC, from traditional methods to machine learning (ML) methods. In paper [1], likelihood-based (LB) techniques for classifying linear digital modulation are examined. In [2] fuzzy method for multi-class classification with three modulation schemes is proposed. Different statistical features were extracted from signals and fed to support vector machines (SVM) for classification. The model provided accuracy of about 77% at 20dB SNR, but their performance at low SNR ($5dB$) was only 55%. Another approach based on SVM is proposed in [3] where the spectral correlation of signals was used as a feature. Classification of six digital modulation schemes like BPSK, QPSK, 2FSK, MSK, 8PSK, and 4PSK was carried out in [3]. Researchers have recently started exploring DL-based AMC techniques since they outperform traditional ML techniques. In [4] two layers of CNN in succession to two layers of dense fully connected layers are used to classify eleven modulation schemes, including digital and analog modulation schemes. [5] adopted the CNN model of [4] and used the amplitude and phase information as input to the model, and achieved a minor performance advancement. A CNN model with Gaussian noise layer is introduced in [6] for performance improvement. Modulation classification based on constellation diagram is proposed in [7]. In [8], a method utilizing two combined convolutional neural networks (CNN) is presented. [9] proposes a Deep Neural Network (DNN) based method, while [10] employs a polar coordinate approach in deep learning to eliminate the impact of carrier phase shift. In [11] and [12], LSTM (Long Short-Term Memory) is integrated with the CNN model to capture time-related and spatial features.For classification purposes, a hybrid architecture is employed, which combines elements of both the residual and inception structures along with the LSTM module. Finally, [13] proposes a hybrid two-stream model for modulation classification using CNN and LSTM to extract spatial and temporal information and improve accuracy. The research efforts discussed in the previous works have a common focus:

developing AMC techniques applicable to both analog and digital modulations while aiming for high accuracy, albeit at the expense of complexity. Given that the radio environment is dynamic and constantly evolving, any AMC model must undergo repeated training to adapt effectively to these changes. Therefore, it is imperative to design a DL architecture that minimizes the training time required. Since the upcoming generation of wireless communication primarily revolves around digital modulation techniques, our research introduces an innovative dual-stream model specifically designed for digital modulations. This novel architecture not only excels in terms of classification accuracy but also manages to outperform state-of-the-art models while maintaining lower complexity.

II. Signal Model

The received signal $r_x(t)$ can be represented as the following using the wireless impulse response model:

$$r_x(t) = \exp\left[j \cdot \Delta f(t)\right] \int_{t=0}^{\tau_0} s\left[\Delta c(t-\tau)\right] h(t,\tau) d\tau + n(t), \quad (1)$$

here, j denotes an imaginary number, $\Delta f(t)$ denotes carrier frequency offset, $\Delta c(t)$ denotes sampling rate offset, τ denotes maximum delay spread, $s(t)$ denotes complex transmitted signal, $h(t,\tau)$ denotes fading channel and $n(t)$ shows AWGN with zero mean and $\sigma_{\bar{n}}^2$ variance. Here $r_x[n]$ is the discrete form of $r_x(t)$ with $f_s = 1/T_S$ sampling rate. $r_x[n]$ contain in-phase $r_x^I \in R$ and quadrature components $r_x^Q \in R$ both and $r_x[n]$ can be represented as

$$r_x(n) = r_x^I[n] + jr_x^Q[n] \quad (2)$$

Let there be N samples in a sampling period, then j^th^ data vector can be represented as

$$r_{xj} = [r_{xj}[0], r_{xj}[1], r_{xj}[2], ..., r_{xj}[N-1]]^T \quad (3)$$

where the T, superscript is the transpose operator. An open source RML2016.10B is used in this work [14]. Table I contains a full description of the RML 2016.10b dataset's requirements. In this study, the dataset is used to extract the digital modulations QAM16, CPFSK, 8PSK, QAM64, GFSK, BPSK and QPSK.

III. Signal Preprocessing

In this work, we make use of two data representations: Inphase/Quadrature (I/Q) and Amplitude/Phase (A/P). The corresponding transformation for different data representations is as follows.

1) **I/Q representation:** Let $X_j{}^I$ be the j^{th} in-phase and and $X_j{}^Q$ be the quadrature component of I/Q vector. The source of the I/Q vector is the raw complex samples in the dataset. The vector I/Q can be represented as

$$X_j^{I/Q} = \begin{bmatrix} X_j^{I^T} \\ X_j^{Q^T} \end{bmatrix}, \quad (4)$$

where $X_j^{I^T}, X_j^{Q^T} \in R^N$.

2) **A/P representation:** Let X_j^A and X_j^P be the amplitude and phase vector, respectively of the A/P vector. The vector A/P can be represented as

$$X_j^{A/P} = \begin{bmatrix} X_j^{A^T} \\ X_j^{P^T} \end{bmatrix}, \quad (5)$$

where $x_j^A[n]$ and $x_j^P[n]$ belonging to $X_j^{A^T}$ and $X_j^{P^T}$ can be computed as

$$x_j^A[n] = \sqrt{\left[r_{xj}^I(n)\right]^2 + \left[r_{xj}^Q(n)\right]^2}, \; x_j^P[n] = \arctan\left(\frac{r_{xj}^Q[n]}{r_{xj}^I[n]}\right). \quad (6)$$

IV. Proposed Method

The proposed model is a two stream architecture in which Convolution and gated recurrent unit (GRU) layers are used. The GRU layer can perform the same tasks as LSTM, but it boasts a relatively simple architecture, faster training convergence, and more efficient memory usage when compared to LSTM. The attention layer is used so that the model can memorize a large sequence of data. Fig.1 depicts the proposed model's architecture.

In both streams of the proposed architecture, similar components are used. The details of components in the single stream are explained below.

- The initial layer is a convolutional layer containing 256 filters of size 2×3. As an activation function the Rectified Linear Unit (ReLU) is used. Following the convolutional layer, a dropout layer with a dropout rate of 0.5 is included. To ensure the output shape is consistent before the next layer, zero padding is applied. The feature map generated by this layer has a size of 260×130 and is passed as input to the subsequent GRU layer.
- The second layer is an GRU layer with 50 memory units, and a dropout rate of 0.5 is added. The output of this layer is then fed to the attention layer, which is based on the approach proposed in [15]. The GRU layer generates a 50×130 feature map, which is a set of feature vectors, and this is provided as input to the attention layer.
- Finally, the third layer is an attention layer that produces a 50×1 feature vector as output.

The diversification of features, as defined in Equation (7), is achieved through a process involving an outer product operation. After obtaining the feature vectors from both streams, where f_{str1} represents the feature vector derived from stream 1 and f_{str2} represents the feature vector obtained from stream 2, these feature vectors are created based on the respective input I/Q vector $X_j^{I/Q}$ and A/P vector $X_j^{A/P}$. This specific step facilitates the interaction of features from both streams in pairs, enhancing feature diversity. here f_{str1} represents the feature vector derived from stream 1, and f_{str2} represents the feature vector obtained from stream 2. These feature vectors are generated based on the respective input I/Q vector $X_j^{I/Q}$ and A/P vector $X_j^{A/P}$.

$$f_{str1} \otimes f_{str2} = f_{str1}\left(X_j^{I/Q}\right) f_{str2}\left(X_j^{A/P}\right)^T. \quad (7)$$

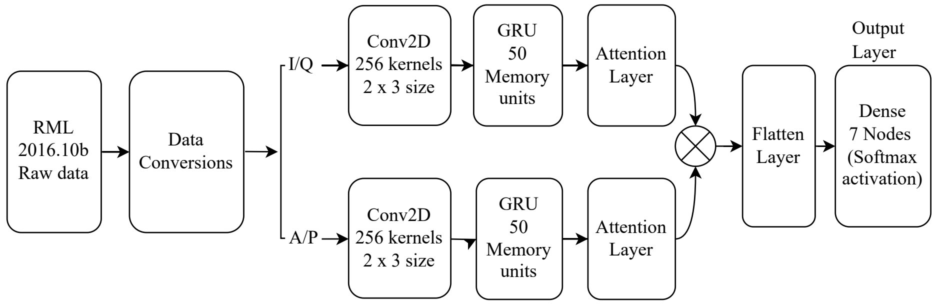


Fig. 1: Proposed Architecture.

TABLE I: RadioML2016.10b Dataset Parameters.

Parameter	Value
Modulation schemes	WBFM, AM-DSB, QAM16, PAM4, QPSK, QAM64, 8PSK, BPSK, CPFSK, GFSK
Length per example	128
Signal format	In-phase and Quadrature (IQ)
Number of examples per SNR for each modulation	6000
SNR range	[-20dB : 2dB : 18dB]
Duration per example	128 μs
Total number of examples in dataset	1200000
Sampling frequency	1 MHz

Following the outer product operation, the resulting 50×50 output is flattened through a flatten layer. Subsequently, a dense softmax layer is utilized to combine the feature vector, which then maps the collected features to various modulation schemes. The model's output is then represented as:

$$\hat{Z} = F\left(X_j^{I/Q}, X_j^{A/P}, \alpha\right), \tag{8}$$

The function F(·) refers to the overall operation of the classifier, with α representing the trained model parameters. Due to its capacity to deliver precise findings with little computation and quicker convergence, the categorical cross-entropy error is used as the selected loss function. The mathematical representation of the loss function is

$$L(\alpha) = -\frac{1}{M}\sum_{k=1}^{M} z_k \log \hat{z_k} + (1 - z_k)\log(1 - \hat{z_k}), \tag{9}$$

where M represents the number of samples within the training batch, where z_k signifies the target output, and $\hat{z_k}$ represents the predicted output. The objective during training is to minimize the loss function, and for this purpose, the Adam optimizer is employed. The choice of using the Adam optimizer is recommended due to its advantages, which include fast computation times and requiring fewer tuning settings, making it an efficient and effective optimization algorithm for this task. The network parameters are updated repetitively according to:

$$\alpha_{t+1} = \alpha_t - \eta \Delta L(\alpha_t), \tag{10}$$

where η is the learning rate, Δ is the gradient operator and $L(\alpha_t)$ is the loss function and subscript t represents the t^{th} iteration. The values of validation loss and α keep updating with each iteration till the validation loss achieves the minimum value.

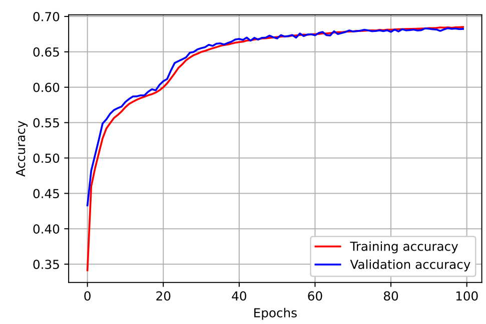


Fig. 2: Accuracy Vs epochs

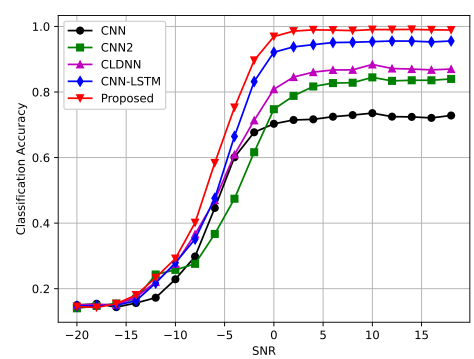
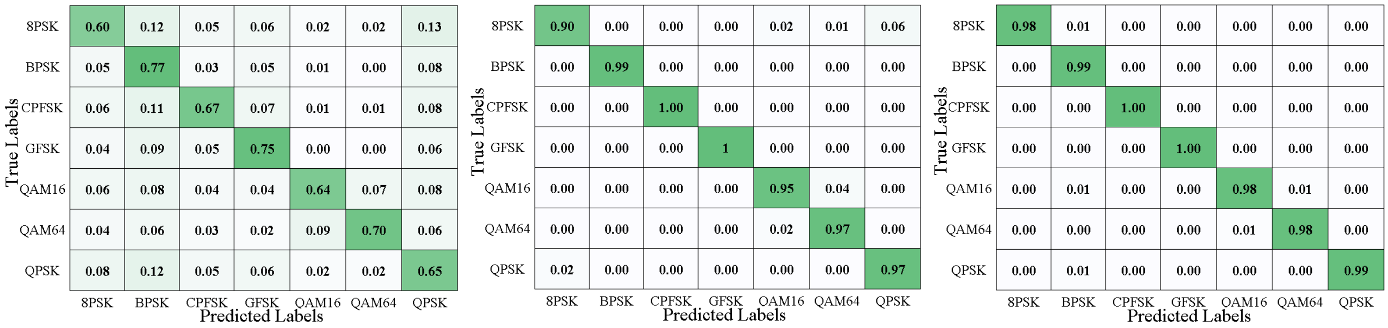


Fig. 3: Comparison of proposed model with DL-based methods at different SNRs.

V. Results and Discussions

The experiments involve dividing the data into two sets: 80% for training and 20% for testing. During training, each epoch processes 100 batches, each containing 1024 examples. The optimization is performed using the Adam optimizer with a learning rate set to 0.001. The training is conducted on a Quadro RTX 6000 GPU with 24GB of memory, running on CUDA version 11.2. The figure labeled as Fig. 2 illustrates the



(a) (b) (c)

Fig. 4: Confusion matrix for proposed model for (a) all SNR values, (b) 0 dB SNR and (c) 18 dB SNR.

variation in accuracy as a function of the number of epochs for the model. It is evident from the plot that the proposed model does not suffer from overfitting, as the accuracy remains stable and does not significantly diverge between the training and testing datasets as the number of epochs increases. This indicates that with an increase in the number of epochs, there is a corresponding improvement in accuracy, which reaches a saturation point at around 80 epochs. The plotted graph displays both the training and validation accuracy. Figure 3 displays the accuracy versus SNR to provide a detailed analysis of the proposed model's performance. The results show that the proposed model achieves an accuracy of over 75% across the SNR range from −4 dB to 18 dB. Furthermore, within the SNR range of 2 dB to 18 dB, the accuracy surpasses 96%. To evaluate the performance of the proposed model, a comparison is made with several recent works. Specifically, models named CNN [4], CNN2 [5], CLDNN [12] and CNN-LSTM [13] were trained with a batch size of 1024 and 100 epochs for digital modulation schemes. The results obtained from these reference models are then compared to the results of our proposed model. The findings indicate that the proposed model exhibits significantly higher classification accuracy than the reference models across a wide SNR range, spanning from −16 dB to 20 dB. This suggests that the proposed model outperforms existing models for digital modulation classification in various SNR conditions. The results clearly demonstrate that the proposed model outperforms the model discussed in [13]. In [13], their model's classification accuracy was compared with various traditional ML algorithms such as KNN, Linear SVM, Random Forest, Gaussian Naive Bayes, and Expert SVM, as well as DL based architectures like CNN and LSTM. Their model showed better performance than these algorithms. Given that the proposed algorithm performs significantly better than the model in [13], it is expected to also outperform traditional ML and existing DL-based algorithms. The confusion matrix for the proposed model across the entire SNR range is depicted in Fig. 4a. This matrix provides a comprehensive view of the classifier's performance. Additionally, confusion matrices for different SNR levels are shown in Fig. 4 to evaluate the model's performance at both low and high SNRs. As demonstrated in Figure 4b, the diagonal elements of the confusion matrix are nearly 1, indicating the outstanding performance of the proposed model. This means that the model can accurately classify signals even in low SNR conditions, such as 0 dB. Furthermore, even at an SNR of −6 dB, the model can achieve an accuracy of over 50. At 18 dB SNR, all diagonal elements are close to 1, as shown in Fig. 4c, suggesting that the proposed model performs exceptionally well at high SNR levels. Table II provides a comparison between the proposed model and the model presented in [13] in terms of accuracy, complexity, training time, and testing time. To ensure a fair comparison, the same dataset used for the proposed model was utilized as input for the model in [13]. From the table, it is evident that the proposed model outperforms the previous work in [13]. As for [4], CNN2 [5] and CLDNN [12], which exhibited poorer performance compared to the proposed architecture, their complexity comparison is not included in Table II. The proposed model not only enhances classification accuracy but also reduces model complexity, thereby reducing the time required for classification.

TABLE II: Comparison of Different Architecture Results.

Model	Trainable Parameters	Accuracy	Training Time (minutes)	Testing Time (seconds)
CNN-LSTM [13]	1149528	64.5	219	8
Proposed Model	145851	68.23	59	2.9

VI. Conclusion

This paper introduces a novel approach, a hybrid dual stream CNN and GRU model enhanced with an attention mechanism. This architecture is designed to capture both spatial and temporal features effectively. The inclusion of an attention layer proves beneficial for processing lengthy data sequences, improving the model's ability to remember critical information. The proposed model demonstrates its effectiveness in classifying digital modulation schemes across a wide range of SNRs. The simulation results affirm that the suggested classifier can learn and categorize radio signals with high accuracy. One notable advantage of this model is its reduced complexity and faster computation time, achieved by employing fewer layers and filters. This efficiency makes the model suitable for real-time applications, allowing it to operate seamlessly in practical scenarios.

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